



FACULTY GUIDE

Faculty Game Overview Guide

What students experience, how the adaptive engine responds, and what faculty should know before assigning a game.

Current architecture: Composer-generated games are self-contained browser HTML files. Official themes and optional faculty images are embedded in the package; Concept Review routing is embedded in the game, while the PDFs stay in the Mastery Quests web library. Remote collection is OFF by default; optional operational builds require explicit faculty configuration.

1. The learning model

Mastery Quests combines retrieval practice, mode-specific feedback, adaptive question selection, staged support after errors, escalating difficulty, checkpoint encounters, and Mastery Report. When warranted, the report connects a meaningful weakness to a targeted one-page Concept Review before the student returns to practice.

- Correct answers move the learner forward and contribute evidence about concepts, skills, and question forms.
- Misses can trigger targeted repair, bridge, and retest questions when the mode and bank support them.
- The selector limits recent repeats and uses the available metadata to keep the next item instructionally relevant.
- Checkpoint questions test readiness at major milestones without confining ordinary concepts to a single stage.

2. The ten modes

Mode	What changes	When to use it
Standard Campaign	A 30-room campaign with checkpoints, milestones, setbacks, and browser-local save/resume.	Use for the complete game experience.
Timed Trial	A ten-minute run with a visible countdown and no checkpoints; the timer governs completion.	Use for fluency practice after students know the material.
Exam Drill	Adaptive nine-question sections with review and revision before checkpoint commitment; correctness stays hidden while a section is active.	Use for section-based review and deliberate answer commitment.
Legendary Mode	A 30-room campaign using mastery-level, multi-step questions with close choices and no normal remediation sequence.	Use as a later challenge after ordinary mastery is stable.
Score Attack	A scored run rewarding correct answers, speed, and streaks while penalizing misses and rapid guessing.	Use for repeat practice and motivation, not first exposure.
Quiz	A fixed run of 1-15 questions chosen by the student while faculty-selected concepts remain fixed.	Use when assigning a specific-length classroom or homework run.

Mode	What changes	When to use it
Unlimited Practice	Adaptive practice repeats in 30-question cycles until the student ends the session and generates a report.	Use for open-ended practice where students decide when to stop.
Trial by Graph	A 10-, 15-, or 20-question run using only audited items in which the graph is required; only supported lengths appear.	Use for concentrated graphical-analysis practice.
Fading Fortune	A 10-, 15-, or 20-question run. Random incorrect choices fade at difficulty-based intervals and value falls 100 -> 75 -> 50 -> 25.	Use for time pressure without a full-run clock.
Risk & Reward	A 10-, 15-, or 20-question run with a 1,000-point bankroll. Students risk 10%, 25%, 50%, or all in before seeing each question.	Use for strategic confidence-and-stakes decisions.

3. What happens after a wrong answer

In supported modes, the engine can route the learner into a short remediation sequence tied to the missed concept or skill. A correct repair can lead to a bridge or retest before ordinary progression resumes. When no appropriate repair route exists, the engine applies the normal room setback instead. Legendary Mode intentionally removes the staged repair sequence.

4. Checkpoints and progression

The visible path uses three checkpoints. The composer maps published easy checkpoint questions to Checkpoint One, medium checkpoint questions to Checkpoint Two, and hard checkpoint questions to the Final Checkpoint. Optional checkpoint focus narrows eligible concepts for a checkpoint; it does not remove those concepts from ordinary question selection.

5. Mastery Report and Concept Reviews

Mastery Report summarizes the evidence from the run and identifies the strongest and weakest signals across concepts or learning objectives and, where applicable, question forms or skills. It is diagnostic evidence, not a replacement for an aligned exam, and low-evidence runs should be interpreted cautiously.

- The production library provides one-page reviews across General Economics, Microeconomics, and Macroeconomics. Generated games embed only the routing they need; the PDFs stay centrally hosted in the Mastery Quests web library.
- When a meaningful weakness maps to an available review, the report can recommend the corresponding one-page Concept Review. A clean run may produce no recommendation. The student opens the PDF, reviews the core idea and worked example, then returns to targeted practice, Repair, or Bridge so later performance creates new evidence.
- A high Score Attack total or fast completion is not the same thing as broad mastery.

6. Saving, Concept Reviews, and privacy

- The game can store the player name or initials, mode state, progress, local results, mastery evidence, and preferences in that browser.
- Standard Campaign and Legendary Mode support browser-local save/resume. Timed Trial, Exam Drill, Quiz, Unlimited Practice, Score Attack, Trial by Graph, Fading Fortune, and Risk & Reward attempts do not resume after leaving the run.

- Public polished games and the free manual template keep progress and activity data in the browser. Remote anonymous telemetry is OFF by default. The Composer offers an explicit optional operational telemetry build; operational telemetry is not research by default. Application telemetry does not intentionally include names, email addresses, LMS account identifiers, university identifiers or free-text responses. Learners choose whether to share local downloads through the course workflow.
- If a course requires a downloaded CSV, screenshot, verification code, or mastery report, the student submits that evidence deliberately through the instructor's chosen LMS or course workflow.
- Clearing browser site data or switching devices can remove locally stored progress.

7. Build or select the experience

Path	Use it when	Faculty work
Faculty Build Composer	You are creating an economics quest from the current library.	Choose concepts and learning-outcome coverage first, then title, modes, checkpoint focus, Appearance, Readiness, and Generate. Use Brief, Standard, Full, or Custom coverage.
Published quest library	A current game already matches the course unit.	Select the game, assign an enabled mode, test the hosted game, and state the required evidence.

8. Instructional use

- Assign a specific mode rather than telling students simply to “play the game.”
- Use Standard Campaign when the full game layer is part of the learning experience.
- Use Exam Drill for section-based review, revision, and checkpoint commitment.
- Use Timed Trial after students know the material and you want fluency under time.
- Use Legendary Mode as a later challenge; Score Attack for repeat practice; Quiz for assigned length; Unlimited Practice for open-ended work; Trial by Graph for graph analysis; Fading Fortune for timed-choice pressure; and Risk & Reward for confidence-and-stakes decisions.
- For low-stakes use, prefer completion or participation over raw speed or a single weakness label.

Current guidance: <https://masteryquests.org/how-to/composer/> | Learning outcomes:

<https://masteryquests.org/how-to/learning-outcomes/> | Manual authoring and validation:

<https://masteryquests.org/downloads/resources/manual-faculty-guide.html>